# Technical design

**Project Virtualisatie**

**V0.3**

| Versie beheer | | | |
|---|---|---|---|
| **Versie** | **Datum** | **Veranderingen** | **Auteur(s)** |
| V0.1 | 13-09-2014 | TD document | Frank |
| V0.2 | 14-09-2014 | Network | Frank |
| V0.3 | 16-09-2014 | System design | Robin & Ahmed |
| V0.4 | 04-10-2014 | Updated archmate | Frank |
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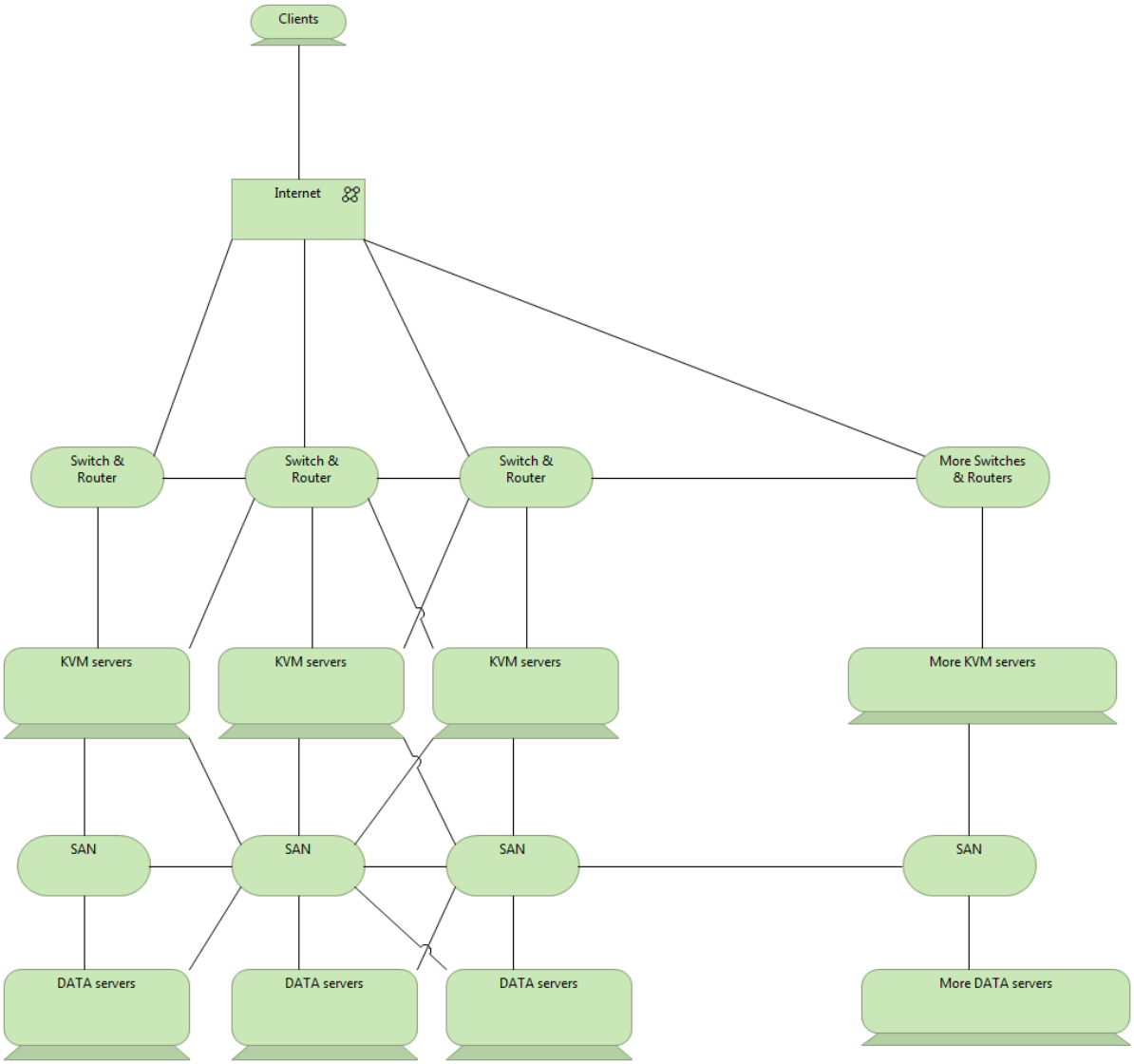
## Network design

This part of the document describes the kind of network we are going to deliver. First of, the most important part of designing the network is making it fast and reliable. Doing so we need to break the network up in smaller parts to insure fast data traffic, this is called subnets.  We will make de subnets as small as possible this will decrease network collisions and therefore increase network speed.

Reliability is also very important to ensure maximum network uptime. We think it's best to use a form of backup where we connect all network devices to switches. And make sure they overlap all there connections. For example: when one router would break all the network traffic would be going to the second router. This will also work with high demand data traffic. when there is a spike of traffic at one side of the network. It will load balance the data to the rest of the network.

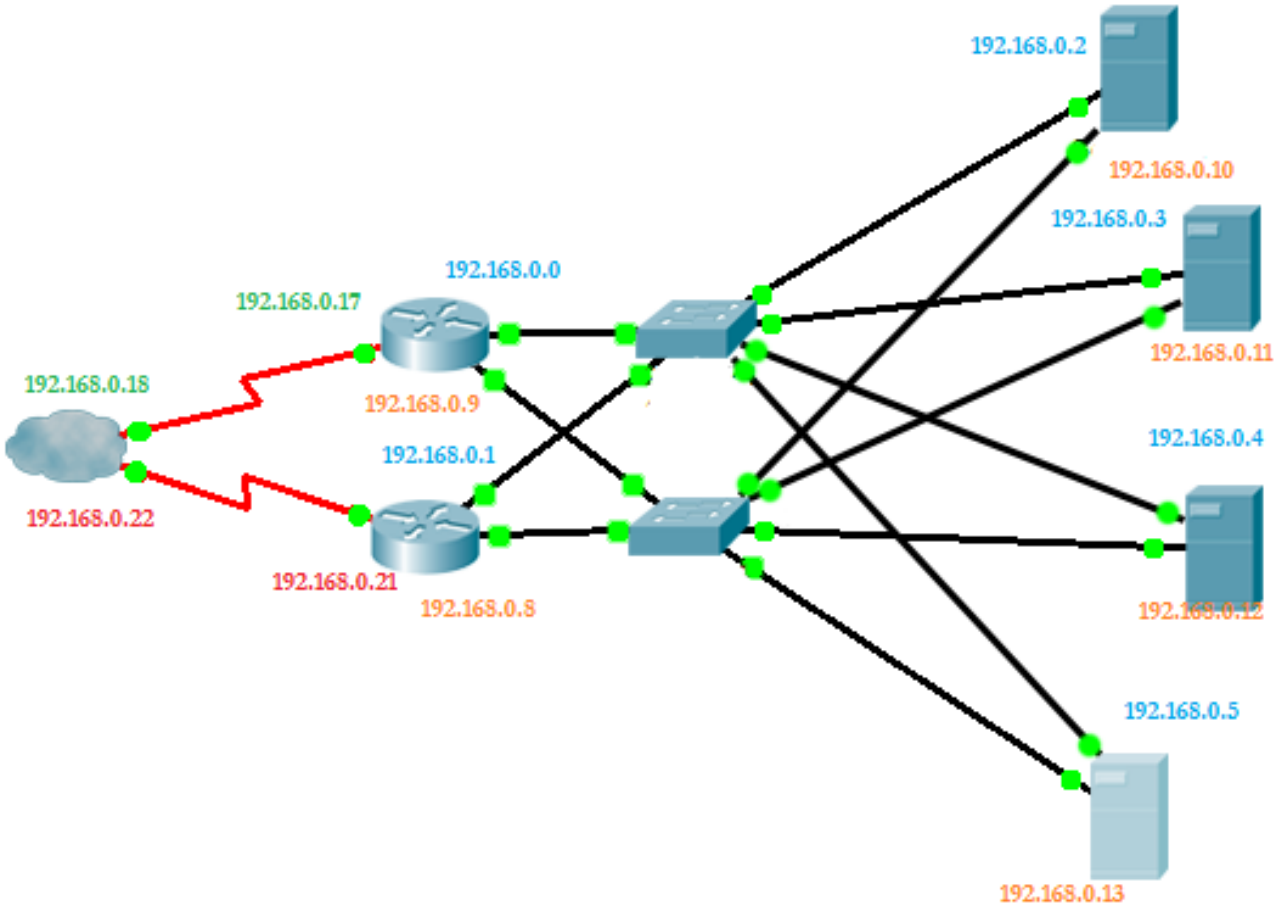
## Network configuration



## Detailed Network example

We plan to use the following network configuration. What is optimized for speed and load balancing.

| | Subnet ID | First adress | Last Adress | Broadcast adress | Subnetmask |
|---|---|---|---|---|---|
| subnet 1 | 192.168.0.0/29 | 192.168.0.1 | 192.168.0.6 | 192.168.0.7 | 255.255.255.248 |
| subnet 2 | 192.168.0.8/29 | 192.168.0.9 | 192.168.0.14 | 192.168.0.15 | 255.255.255.248 |
| subnet 3 | 192.168.0.16/30 | 192.168.0.17 | 192.168.0.18 | 192.168.0.19 | 255.255.255.252 |
| subnet 4 | 192.168.0.20/30 | 192.168.0.21 | 192.168.0.22 | 192.168.0.23 | 255.255.255.252 |



## System design

# System configuration

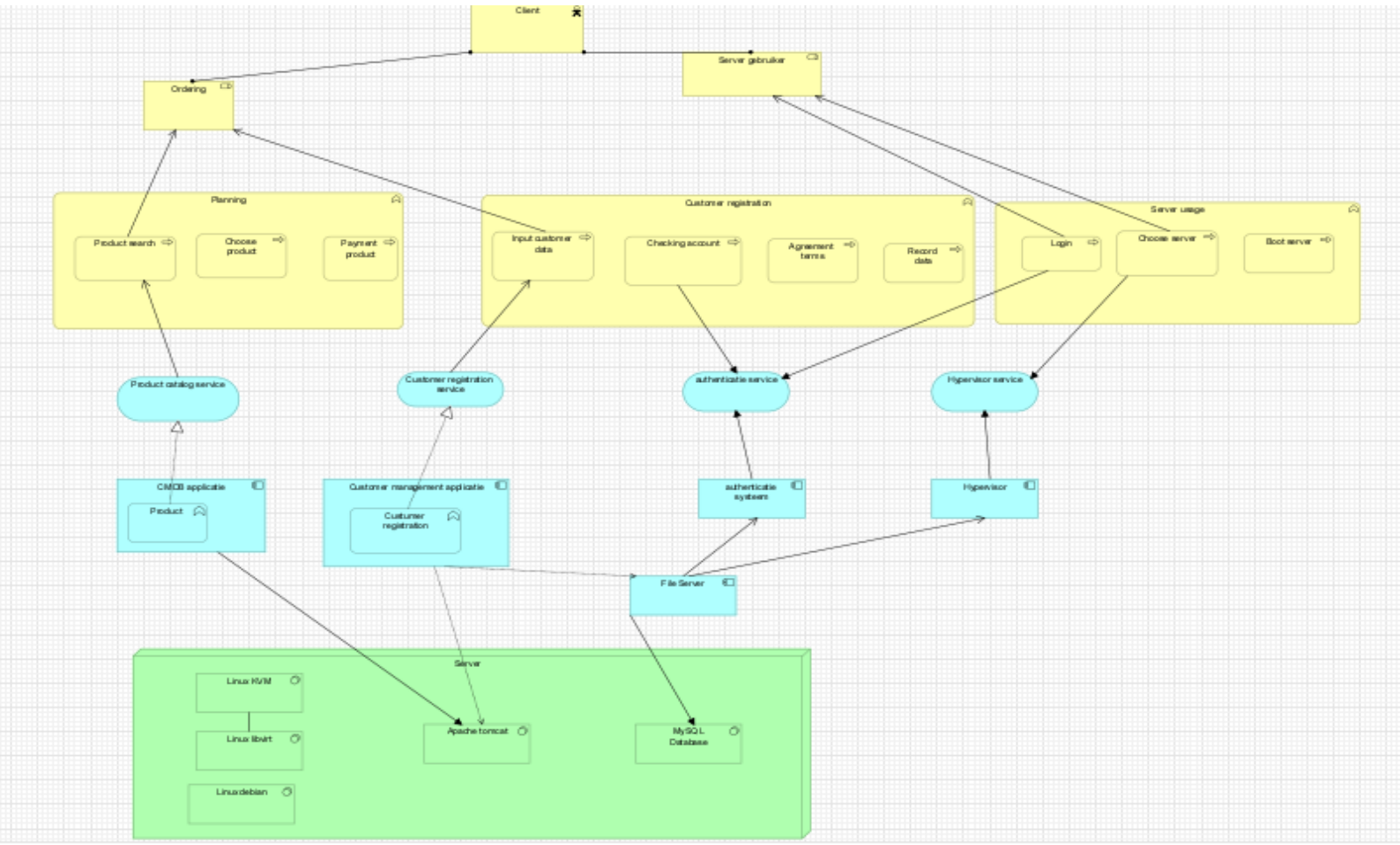
**Routers**

Cisco 5760

We chose the cisco 5760 because it is designed for mid-to-large campus deployments and offers:

- Support high availability and application visibility for wireless clients
- Wire-speed 60 Gbps throughput with advanced network services per controller
- Support for up to 1000 access points and 12,000 clients per controller
- High resiliency with N+1 clustering, Multiple LAG, and redundant power supplies
- Cisco IOS-based wireless controller with features such as Flexible NetFlow, Advanced QoS, downloadable access control lists, and more

**Server**

Dell PowerEdge r620

Designed for extraordinary performance on a wide range of applications, the Dell™ PowerEdge™R620 rack server is an impressive general-purpose platform that is perfect for data centers with space constraints. With its hyper-dense memory — up to 768GB in a 1U form factor — and powered by the Intel®